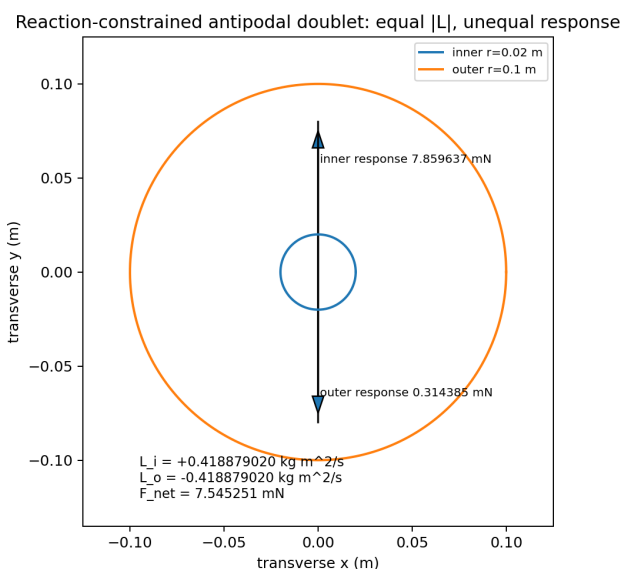


K-FIELD

K-Field Counter-Rotating System Is an Antipodal Doublet

Antipodal parity, unequal specific areal inertia, residual lateral force, Cell3 inversion eigenspaces, and multiscale geometric response



Foundational geometry: an equal-and-opposite angular-momentum doublet can retain a common-direction lateral response when the two channels have unequal specific areal inertia.

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Krytronx, LLC

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This paper is a standalone foundational education and archival reference. It develops the counter-rotating antipodal-doublet principle from its mechanical definitions, the adopted K-Field response law, the Cell3 metric geometry, the antipodal sphere theorem, and the Mass-Bridge counter-rotor relations. Every numerical table and plotted quantity is generated from the three source ledgers identified in the reproducibility record or from direct algebraic consequences of those ledgers.

Central result

Antipodal counter-rotation creates a pure odd angular-momentum mode. Unequal reciprocal specific areal inertia mixes that odd mode into an even common-direction lateral-force mode. Zero total angular momentum and zero lateral response are therefore different null conditions.

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Abstract

A counter-rotating pair can have exactly zero total angular momentum while retaining a nonzero lateral response. The reason is that angular momentum and the adopted K-Field rotor response are different functionals of the same motion. Angular momentum weights angular speed by moment of inertia, $I \omega = \beta m r^2 \omega$. The lateral response weights angular speed by mass, $C_K m \omega$. Under a reaction-only internal spin transfer, the two rotors carry equal and opposite angular momentum, but their force contributions are proportional to the reciprocals of βr^2 . Unless the specific areal inertias are equal, the force terms do not cancel.

The resulting geometry is an antipodal doublet. In the parity basis, equal and opposite angular momentum is a pure odd eigenmode. The unequal geometric response operator contains an odd contrast term. The product of the odd state and the odd contrast generates an even common-mode component, which is the net lateral force. The normalized conversion coefficient is $A = (s-1)/(s+1) = \tanh(\xi)$, where s is the ratio of reciprocal specific areal inertias and ξ is one-half its natural logarithm.

Cell3 supplies the higher-dimensional completion. The centered 27-node registry decomposes exactly into one inversion-fixed center and 13 antipodal pairs. Its inversion operator therefore has a 14-dimensional even eigenspace and a 13-dimensional odd eigenspace. The 13 odd channels form the complete finite Cell3 basis for antisymmetric counterflow states. The Cell3 Gram metric further resolves three principal metric dimensions, while the spatial cross-product response has one null direction parallel to the transformed Earth-CMB field vector and two transverse helical eigenmodes.

For an arbitrary multirotor system with signed angular momenta h_j and reciprocal specific areal inertias a_j , the net response is $F_{\text{net}}/C_K = \sum a_j h_j$. When total angular momentum vanishes, this becomes $N \text{Cov}(a, h)$. The residual force is therefore the geometric covariance remaining after the uniform angular-momentum mode has been removed.

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1. Scope, notation, and evidentiary layers

The objective is to identify the exact geometric principle by which counter-rotating masses at different mass scales and different velocities can carry equal and opposite angular momentum yet produce a residual linear force. The paper treats the supplied K-Field force law and the supplied Cell3 and Mass-Bridge ledgers as authoritative inputs. It separates four evidentiary layers so that students can distinguish recorded quantities from derived structure.

Layer	Content
Authoritative input	Numerical constants, Cell3 geometry, field vector, rotor states, balance conditions, and finite-domain antipodal certificate copied from the source ledgers.
Direct algebraic consequence	Equations obtained by substitution, change of basis, normalization, or exact summation of source quantities.
Computed spectral result	Eigenvalues, eigenvectors, parity dimensions, scale coordinates, and generated surfaces computed from the authoritative matrices and arrays.
Structural hypothesis	A clearly identified extension that connects the finite mathematical structure to broader K-Field architecture, such as the possible 6+1 relation to the seven-sheet organization.

1.1 Symbols and units

Symbol	Meaning	Unit
m	rotor mass	kg
r	effective rotor radius	m
beta	dimensionless axial inertia shape factor	1
$I = \beta m r^2$	axial moment of inertia	kg m ²
omega	signed angular speed	rad/s, with rad dimensionless
$L = I \omega$	signed angular momentum	kg m ² /s
C_K	maximal-orientation rotor response coefficient	N s/kg
$a = 1/(\beta r^2)$	reciprocal specific areal inertia	m ⁻²
g	transformed Earth-CMB phase vector	m/s
G	Cell3 Gram metric	native ²
A	normalized parity-conversion coefficient	1
xi	logarithmic scale coordinate	1

1.2 Reproducible structural coverage

The three source archives were indexed as trees of dictionaries, arrays, and scalar leaves. A fixed-seed protocol selected 40 entry paths from each archive, for 120 total entry points. Tree-distance expansion and deduplication produced a 475-node structural registry. The complete list of entry paths and covered structural paths is preserved in the machine-readable companion. This coverage protocol is not a source of physical evidence; it is a reproducibility mechanism ensuring that the analysis is not dependent on a single sequential reading order.

2. Authoritative Cell3 base geometry

Cell3 is defined by three nonorthogonal primitive vectors a, b, and c. A canonical Cartesian reconstruction places a on the x axis, b in the x-y plane, and c above that plane. This is a metric-preserving coordinate realization: all dot products, lengths, included angles, face areas, and volume are unchanged.

$$\begin{aligned}
 a &= (|a|, 0, 0) \\
 b &= (|b| \cos(\theta_{ab}), |b| \sin(\theta_{ab}), 0) \\
 c_x &= |c| \cos(\theta_{ac}) \\
 c_y &= |c| [\cos(\theta_{bc}) - \cos(\theta_{ab})\cos(\theta_{ac})] / \sin(\theta_{ab}) \\
 c_z &= \sqrt{|c|^2 - c_x^2 - c_y^2}
 \end{aligned}$$

Quantity	Native value	Interpretation
a	0.0207970587	shortest primitive edge
b	0.0364942052	intermediate primitive edge
c	0.049308566	longest primitive edge
theta_ab	83.933483 deg	included angle
theta_ac	80.448130 deg	included angle
theta_bc	50.363232 deg	included angle
A_ab	0.000754721788096512 native^2	face area
A_ac	0.00101125575567906 native^2	face area
A_bc	0.00138578444090954 native^2	largest face area
V	2.84206694521122e-05 native^3	primitive volume

Cell3 base cell: metric-preserving canonical embedding

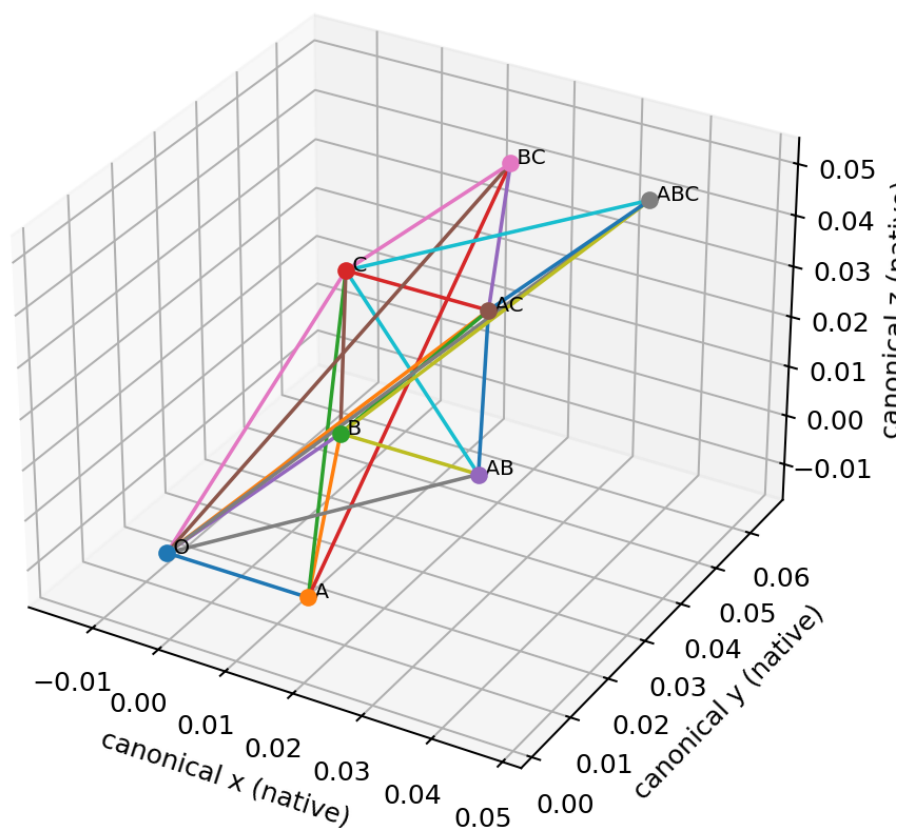


Figure 1. Dimensionally correct Cell3 base cell in the canonical metric-preserving embedding. All eight vertices are labeled, and every plotted edge is an actual primitive-cell edge.

2.1 Complete centered 27-node registry

The centered half-step registry uses addresses (i,j,k) with each coordinate in $\{-1,0,+1\}$. The physical point is one-half the corresponding signed combination of the primitive vectors. The registry contains eight corners, twelve edge centers, six face centers, and one body center. Every nonzero address has exactly one global-sign partner.

Centered Cell3 registry: all 27 half-step nodes

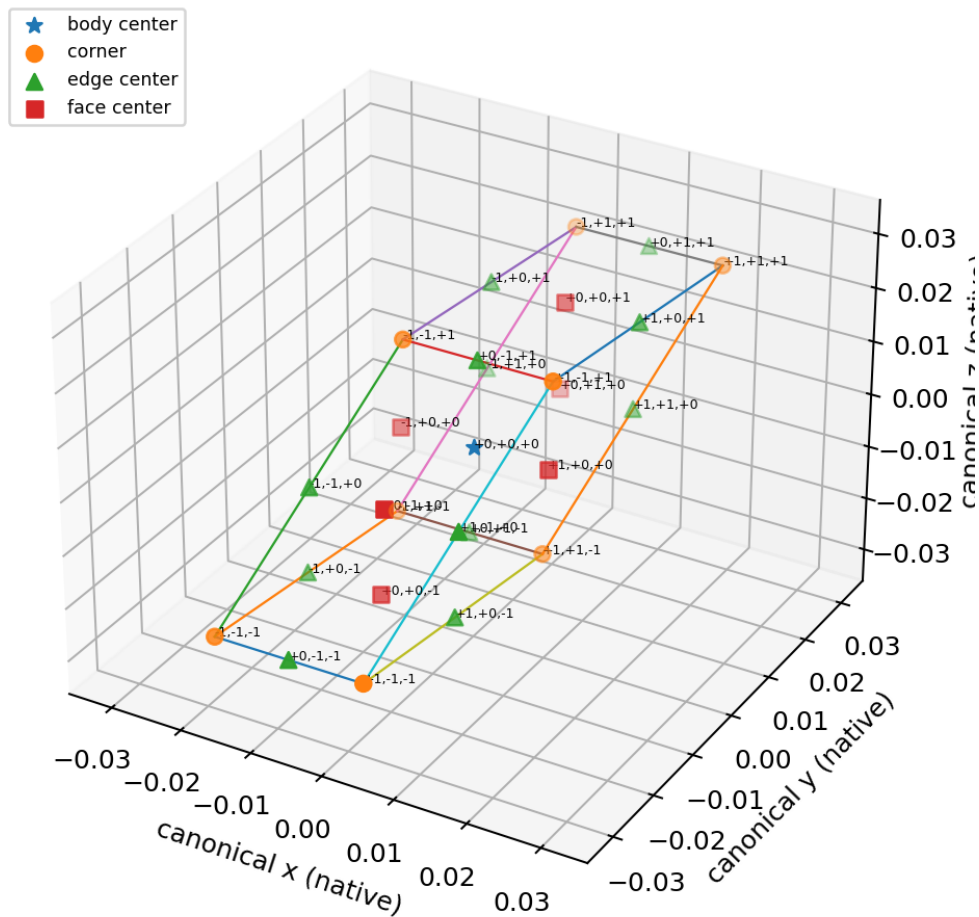


Figure 2. All 27 centered half-step nodes in canonical coordinates. The plot is dimensionally correct and uses equal three-dimensional axis spans. Each node is labeled by its signed address.

Locked study coordinates: 27 nodes and central plane family

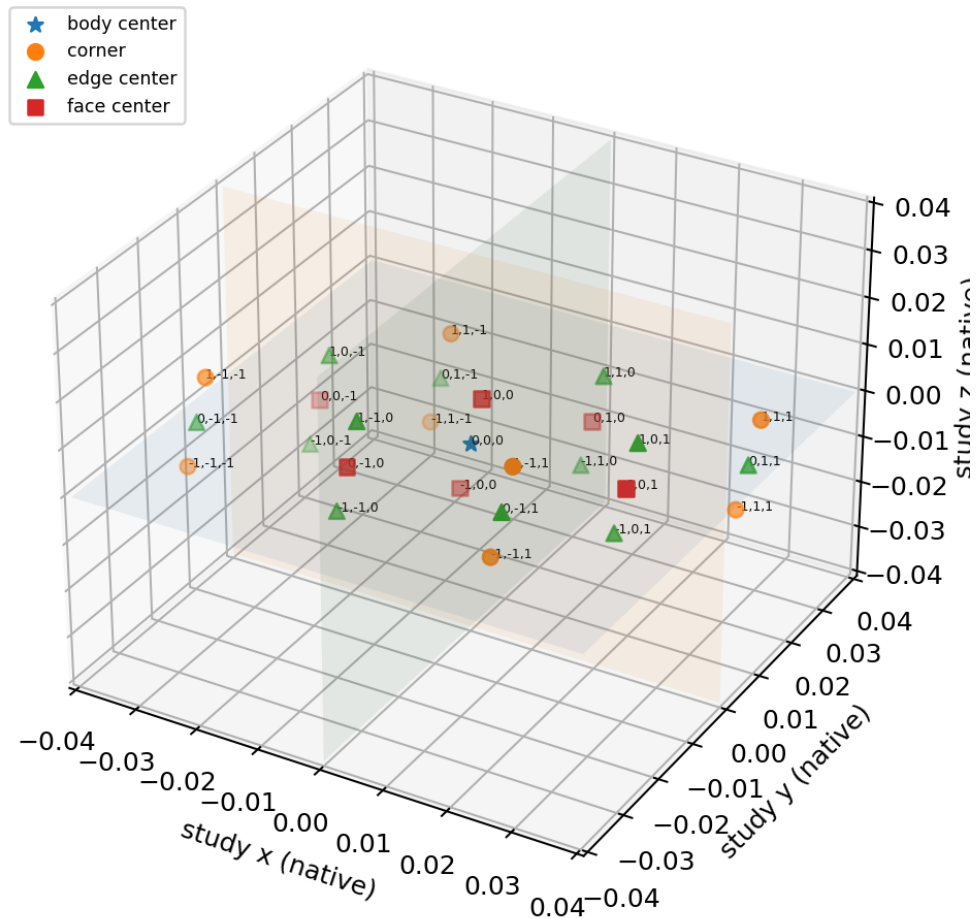


Figure 3. The same 27-node registry in the locked study-coordinate frame used by the antipodal theorem. The three origin-centered coordinate planes show the plane-family structure without changing any metric invariant.

2.2 Complete node coordinate ledger

Address	Kind	x native	y native	z native	x nm	y nm	z nm
-1,-1,-1	corner	-0.0164180796	-0.03352610771	-0.01882857359	-0.2638156015	-0.5387177117	-0.3025488722
-1,-1,+0	edge center	-0.01232693741	-0.01814491652	0	-0.1980766622	-0.2915634583	0
-1,-1,+1	corner	-0.008235795218	-0.002763725333	0.01882857359	-0.1323377229	-0.04440920491	0.3025488722
-1,+0,-1	edge center	-0.01448967154	-0.01538119119	-0.01882857359	-0.2328287782	-0.2471542534	-0.3025488722
-1,+0,+0	face center	-0.01039852935	0	0	-0.1670898389	0	0
-1,+0,+1	edge center	-0.006307387158	0.01538119119	0.01882857359	-0.1013508996	0.2471542534	0.3025488722
-1,+1,-1	corner	-0.01256126348	0.002763725333	-0.01882857359	-0.2018419548	0.04440920491	-0.3025488722
-1,+1,+0	edge center	-0.00847012129	0.01814491652	0	-0.1361030155	0.2915634583	0
-1,+1,+1	corner	-0.004378979098	0.03352610771	0.01882857359	-0.07036407624	0.5387177117	0.3025488722
+0,-1,-1	edge center	-0.006019550252	-0.03352610771	-0.01882857359	-0.09672576264	-0.5387177117	-0.3025488722
+0,-1,+0	face center	-0.00192840806	-0.01814491652	0	-0.03098682335	-0.2915634583	0
+0,-1,+1	edge center	0.002162734132	-0.002763725333	0.01882857359	0.03475211595	-0.04440920491	0.3025488722
+0,+0,-1	face center	-0.004091142192	-0.01538119119	-0.01882857359	-0.06573893929	-0.2471542534	-0.3025488722
+0,+0,+0	body center	0	0	0	0	0	0
+0,+0,+1	face center	0.004091142192	0.01538119119	0.01882857359	0.06573893929	0.2471542534	0.3025488722
+0,+1,-1	edge center	-0.002162734132	0.002763725333	-0.01882857359	-0.03475211595	0.04440920491	-0.3025488722
+0,+1,+0	face center	0.00192840806	0.01814491652	0	0.03098682335	0.2915634583	0
+0,+1,+1	edge center	0.006019550252	0.03352610771	0.01882857359	0.09672576264	0.5387177117	0.3025488722
+1,-1,-1	corner	0.004378979098	-0.03352610771	-0.01882857359	0.07036407624	-0.5387177117	-0.3025488722
+1,-1,+0	edge center	0.00847012129	-0.01814491652	0	0.1361030155	-0.2915634583	0
+1,-1,+1	corner	0.01256126348	-0.002763725333	0.01882857359	0.2018419548	-0.04440920491	0.3025488722
+1,+0,-1	edge center	0.006307387158	-0.01538119119	-0.01882857359	0.1013508996	-0.2471542534	-0.3025488722
+1,+0,+0	face center	0.01039852935	0	0	0.1670898389	0	0
+1,+0,+1	edge center	0.01448967154	0.01538119119	0.01882857359	0.2328287782	0.2471542534	0.3025488722
+1,+1,-1	corner	0.008235795218	0.002763725333	-0.01882857359	0.1323377229	0.04440920491	-0.3025488722
+1,+1,+0	edge center	0.01232693741	0.01814491652	0	0.1980766622	0.2915634583	0
+1,+1,+1	corner	0.0164180796	0.03352610771	0.01882857359	0.2638156015	0.5387177117	0.3025488722

2.3 Dimensionally correct face planes

Each face is plotted in its own orthonormal in-plane coordinate system. The local horizontal axis follows the first named edge; the local vertical axis is the component of the second edge perpendicular to the first. Equal plot scales therefore represent equal native lengths. This prevents the visual distortion that occurs when independent axis ranges are stretched to fill a frame.

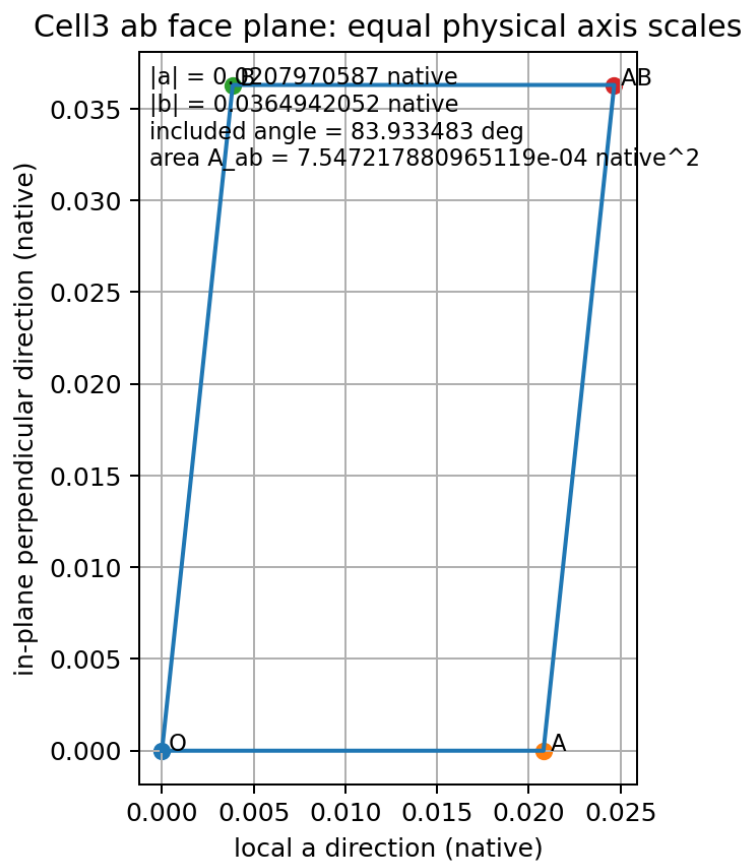


Figure 4. The ab face plane with all four vertices, actual edge lengths, included angle, and physical area. Equal local x and y scales preserve the true plane geometry.

Cell3 ac face plane: equal physical axis scales

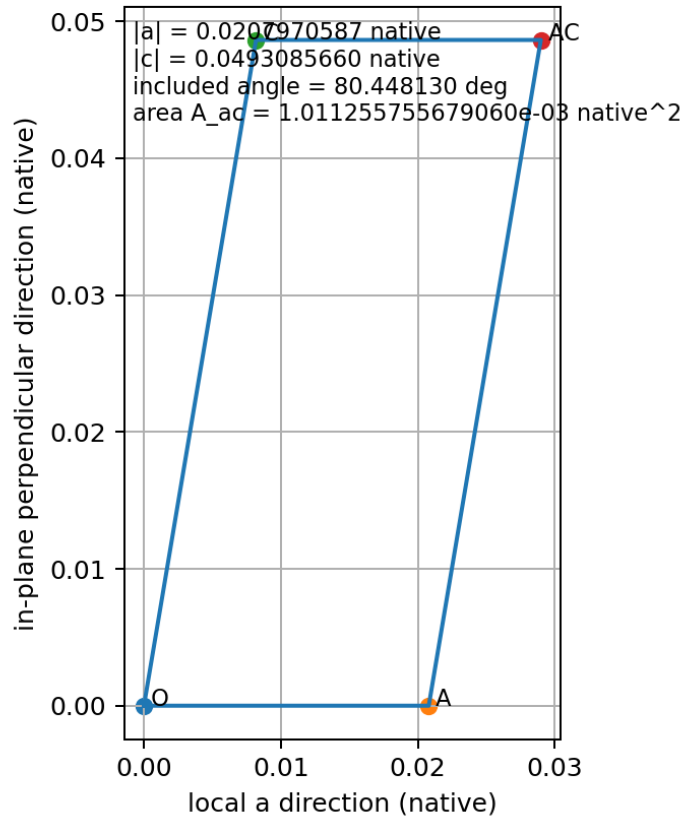


Figure 5. The ac face plane with all four vertices, actual edge lengths, included angle, and physical area. Equal local x and y scales preserve the true plane geometry.

Cell3 bc face plane: equal physical axis scales

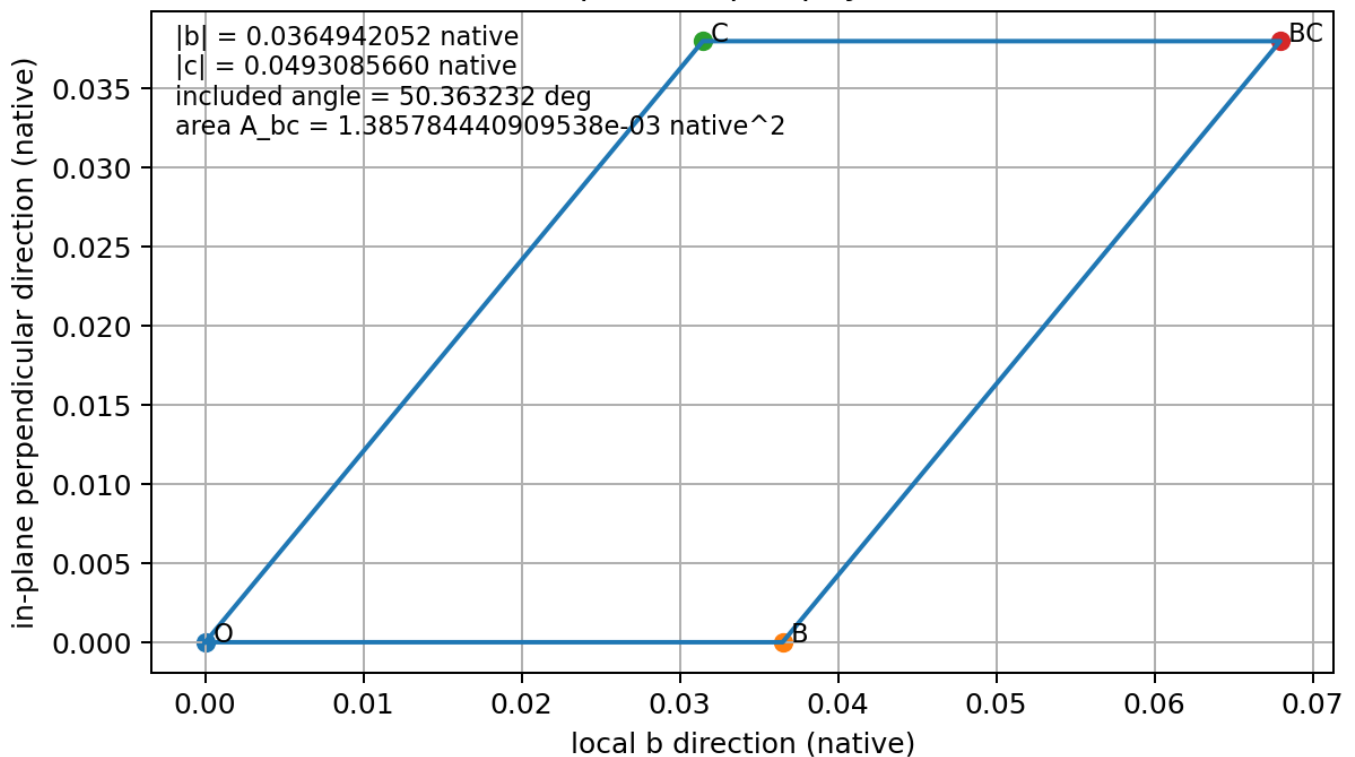


Figure 6. The bc face plane with all four vertices, actual edge lengths, included angle, and physical area. Equal local x and y scales preserve the true plane geometry.

3. Antipodal sphere geometry

For an integer lattice address n , the Cell3 center is $p(n)=B n$ and its squared radius is $n^T G n$. Global sign reversal sends $p(n)$ to $-p(n)$ without changing its radius. The antipodal sphere theorem states that, under antipodal radial injectivity, any occupied origin-centered sphere contains exactly one pair $\{p,-p\}$. The associated circle result follows because every origin-centered plane containing p also contains $-p$.

$$\begin{aligned} p(n) &= B n \\ p(-n) &= -p(n) \\ ||p(n)||^2 &= n^T G n \\ n^T G n = m^T G m &\Rightarrow m = \pm n \end{aligned}$$

Certificate quantity	Value
Index bounds	[-100, 100]
Radius range in x-cell steps	[1, 100]
Nonzero centers in range	6,762,624
Distinct occupied shells	3,381,312
Shell multiplicity distribution	{2': 3381312}
Non-antipodal equal-radius shells	0
Closest-shell squared-radius gap	2.4537351100901747111117945716E-12
Separation / uncertainty ratio	5061.71507595729777926373911065607038882486638899433480645030602887123 67898109849

3.1 Exact inversion dimension count

The finite 27-node cell provides the simplest complete inversion representation. The body center is fixed by inversion. The remaining 26 nodes form 13 disjoint antipodal pairs. Therefore $27=1+2(13)$.

$$\begin{aligned} \dim(V_{\text{even}}) &= 1 + 13 = 14 \\ \dim(V_{\text{odd}}) &= 13 \end{aligned}$$

For each nonzero pair $\{p,-p\}$, the symmetric combination has inversion eigenvalue +1 and the antisymmetric combination has inversion eigenvalue -1. The center contributes only to the even sector. The complete odd Cell3 state space is therefore exactly 13-dimensional.

Cell3 inversion decomposition: 13 antipodal pairs and one fixed center

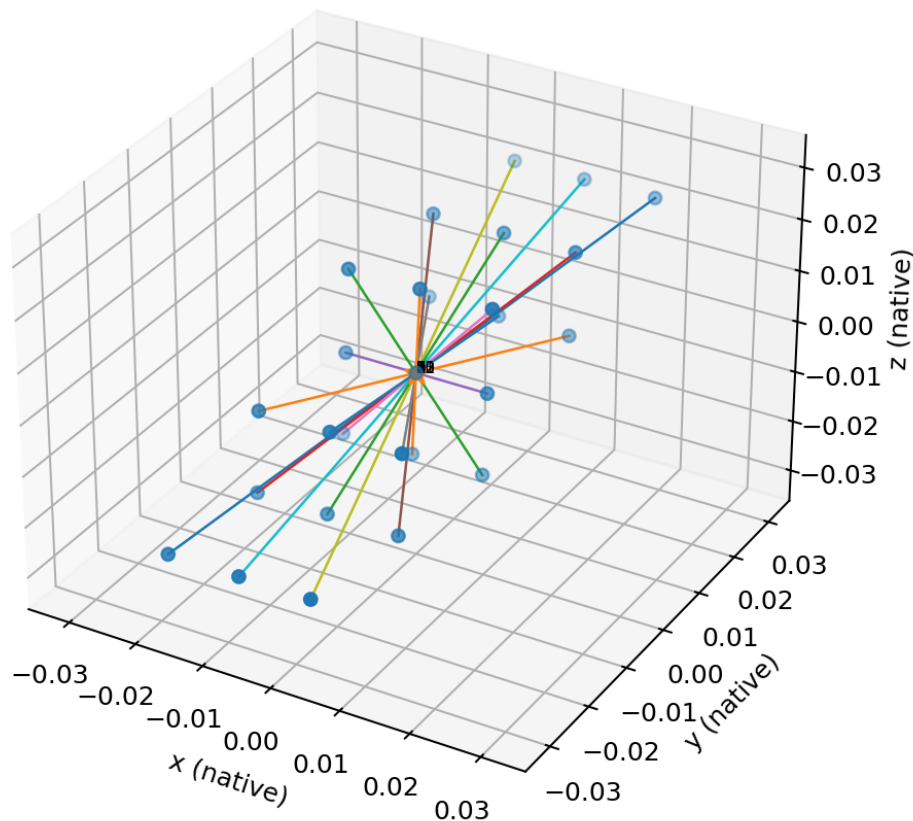


Figure 7. The centered Cell3 registry decomposed into 13 antipodal lines and one inversion-fixed center. This is the finite geometric origin of the 14-even / 13-odd parity split.

4. Mechanical model of a counter-rotating pair

4.1 Inertia, energy, and angular momentum

An axisymmetric rotor is represented by the shape factor beta. A thin ring has beta=1. Other axisymmetric mass distributions have other values, but the derivation depends only on the product beta r².

$$\begin{aligned} I &= \beta m r^2 \\ E &= (1/2) I \omega^2 \\ L &= I \omega \end{aligned}$$

Energy is even under reversal because it contains omega squared. Angular momentum is odd because it contains one signed factor of omega. A counter-rotating pair can therefore store nonzero positive energy while its signed angular momenta cancel.

4.2 Adopted axisymmetric K-Field rotor response

The supplied rotor ledger adopts the maximal-orientation force magnitude $F_K = C_K M |\omega|$ and the vector law $F_K = (\kappa_F M/2)(\Omega \text{ cross } g)$. The vector response is zero when the spin axis is parallel to g and maximal when the axis is perpendicular to g.

$$\begin{aligned} F_{K_vector} &= (\kappa_F M/2) (\Omega \text{ cross } g) \\ F_{K_max} &= C_K M |\omega| \\ C_K &= 7.5054e-6 \text{ N s/kg} \end{aligned}$$

Field quantity	Value
g_x	-8.80221981635000e+05 m/s
g_y	-1.43911436562000e+05 m/s
g_z	2.66335026949000e+05 m/s
$ g $	9.30825217270519e+05 m/s
$\hat{g}$	[-0.945636157361632, -0.154606293310354, 0.286127859459997]
Maximal orientation	$\hat{n} \cdot \hat{g} = 0$
Positive force direction	$\hat{n} \times \hat{g}$ for positive inner spin

Spatial eigenstructure: one null axis and a two-dimensional transverse force plane

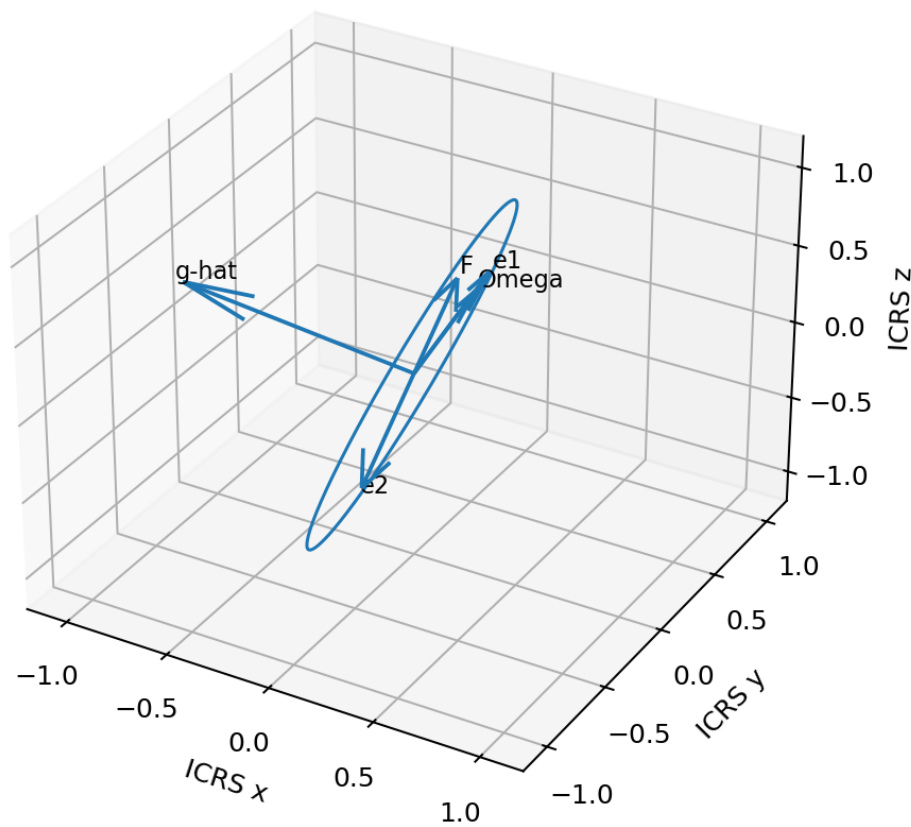


Figure 15. Spatial response eigenstructure. The direction parallel to $\hat{g}$ is the zero mode. The active force space is the two-dimensional plane perpendicular to $\hat{g}$.

5. Central derivation: zero angular momentum with residual force

Let the inner and outer rotors receive equal and opposite angular momentum through an internal reaction. Write $L_i = +L$ and $L_o = -L$. The angular speeds are determined by their different inertias.

$$\begin{aligned}\omega_i &= L / (\beta_i m_i r_i^2) \\ \omega_o &= -L / (\beta_o m_o r_o^2)\end{aligned}$$

The response law weights each angular speed by its mass. Substitution cancels the mass of each rotor separately.

$$\begin{aligned}F_i / C_K &= m_i \omega_i = L / (\beta_i r_i^2) \\ F_o / C_K &= m_o \omega_o = -L / (\beta_o r_o^2)\end{aligned}$$

Residual-force theorem

For a reaction-constrained pair with nonzero transferred angular momentum L , the net lateral response is $F_{\text{net}} = C_K L [1/(\beta_i r_i^2) - 1/(\beta_o r_o^2)]$. Mass affects the speed and energy required to realize L , but it does not appear in the force at fixed L .

$$F_{\text{net}} = C_K L \left[\frac{1}{\beta_i r_i^2} - \frac{1}{\beta_o r_o^2} \right]$$

5.1 Specific areal inertia is the controlling geometric variable

The quantity $I/m = \beta r^2$ is the moment of inertia per unit mass. It has dimensions of area and will be called specific areal inertia. Its reciprocal $a = 1/(\beta r^2)$ is the force weight per unit angular momentum.

$$\begin{aligned} \text{specific areal inertia} &= I/m = \beta r^2 \text{ [m}^2\text{]} \\ \text{reciprocal response weight } a &= 1/(\beta r^2) \text{ [m}^{-2}\text{]} \end{aligned}$$

Two rotors with very different masses and speeds can cancel exactly if their specific areal inertias are equal. Conversely, changing mass alone cannot create a residual at fixed transferred angular momentum when βr^2 remains unchanged.

5.2 Double-null compatibility theorem

For a nonzero reaction-constrained angular-momentum doublet, simultaneous cancellation of angular momentum and lateral force occurs if and only if the specific areal inertias are equal.

$$\begin{aligned} L_i + L_o &= 0 \text{ and } F_{\text{net}} = 0 \\ \Leftrightarrow \frac{1}{\beta_i r_i^2} &= \frac{1}{\beta_o r_o^2} \\ \Leftrightarrow \beta_i r_i^2 &= \beta_o r_o^2 \end{aligned}$$

6. Antipodal parity conversion

The two-rotor system is most transparent in the exchange-parity basis. Let P exchange the inner and outer channels. Its normalized eigenvectors are $e_{\text{plus}} = (1,1)/\sqrt{2}$ and $e_{\text{minus}} = (1,-1)/\sqrt{2}$. The equal-and-opposite angular-momentum state is purely odd.

$$\begin{aligned} P e_{\text{plus}} &= +e_{\text{plus}} \\ P e_{\text{minus}} &= -e_{\text{minus}} \\ h &= (L, -L)^T = \sqrt{2} L e_{\text{minus}} \end{aligned}$$

The geometric response operator in the physical channel basis is diagonal, $D = \text{diag}(a_i, a_o)$. Define the mean and contrast $a_{\text{bar}} = (a_i + a_o)/2$ and $\delta a = (a_i - a_o)/2$. In the parity basis the operator is not diagonal unless $a_i = a_o$.

$$D_{\text{pm}} = \begin{bmatrix} a_{\text{bar}} & \delta a \\ \delta a & a_{\text{bar}} \end{bmatrix}$$

Applying D to the pure odd state produces both an odd output and an even output. The even output is proportional to $\delta a L$ and is the common-mode lateral force.

$$D_{\text{pm}} (0, \sqrt{2} L)^T = (\sqrt{2} \delta a L, \sqrt{2} a_{\text{bar}} L)^T$$

Parity rule

Odd angular momentum multiplied by odd geometric contrast produces an even common-direction response. Antipodal symmetry supplies the sign structure; unequal geometry supplies the parity mixing.

Antipodal parity conversion by unequal reciprocal areal inertia

Odd input state

$$\begin{aligned} D_{pm} &= [[a_{\bar{a}}, \delta a], \\ h &= (L, -L) \xrightarrow{[\delta a, a_{\bar{a}}]} \\ &= \sqrt{2} L e_- \end{aligned}$$

Output response

$$\begin{aligned} \text{even: } &\sqrt{2} \delta a L \\ \text{odd: } &\sqrt{2} a_{\bar{a}} L \end{aligned}$$

odd angular momentum x odd geometric contrast -> even common-mode force

Figure 14. Parity-basis representation of the two-channel response. When δa is zero, even and odd sectors decouple. When δa is nonzero, the odd input develops an even output.

6.1 What antipodal symmetry does and does not do

Antipodal symmetry alone does not guarantee a residual force. It guarantees that the two rotational channels can be organized as equal-magnitude opposite-sign states. If the response weights are also antipodally identical, the pair cancels. The residual requires a second odd object: a geometric contrast that reverses sign when the channels are exchanged.

This distinction prevents a common conceptual error. Counter-rotation is the carrier of the odd state, not the entire force mechanism. The force is the even projection created when the odd state is evaluated through an asymmetrical response metric.

7. Scale as an eigen-coordinate

Define the response-weight ratio $s = a_i/a_o = (\beta_o r_o^2)/(\beta_i r_i^2)$. A normalized force asymmetry is obtained by dividing the difference by the sum.

$$A = (a_i - a_o)/(a_i + a_o) = (s - 1)/(s + 1)$$

Introduce $\xi = (1/2)\ln(s)$. Then $s = \exp(2\xi)$ and the asymmetry is exactly a hyperbolic tangent.

$$\begin{aligned} \xi &= (1/2) \ln(s) \\ A &= \tanh(\xi) \end{aligned}$$

This coordinate converts multiplicative scale ratios into additive displacements. Exchanging the rotors sends s to $1/s$, ξ to $-\xi$, and A to $-A$. The coordinate ξ is therefore a signed scale eigen-coordinate for antipodal response. Cascaded scale transformations add in ξ even when they multiply in ordinary radius coordinates.

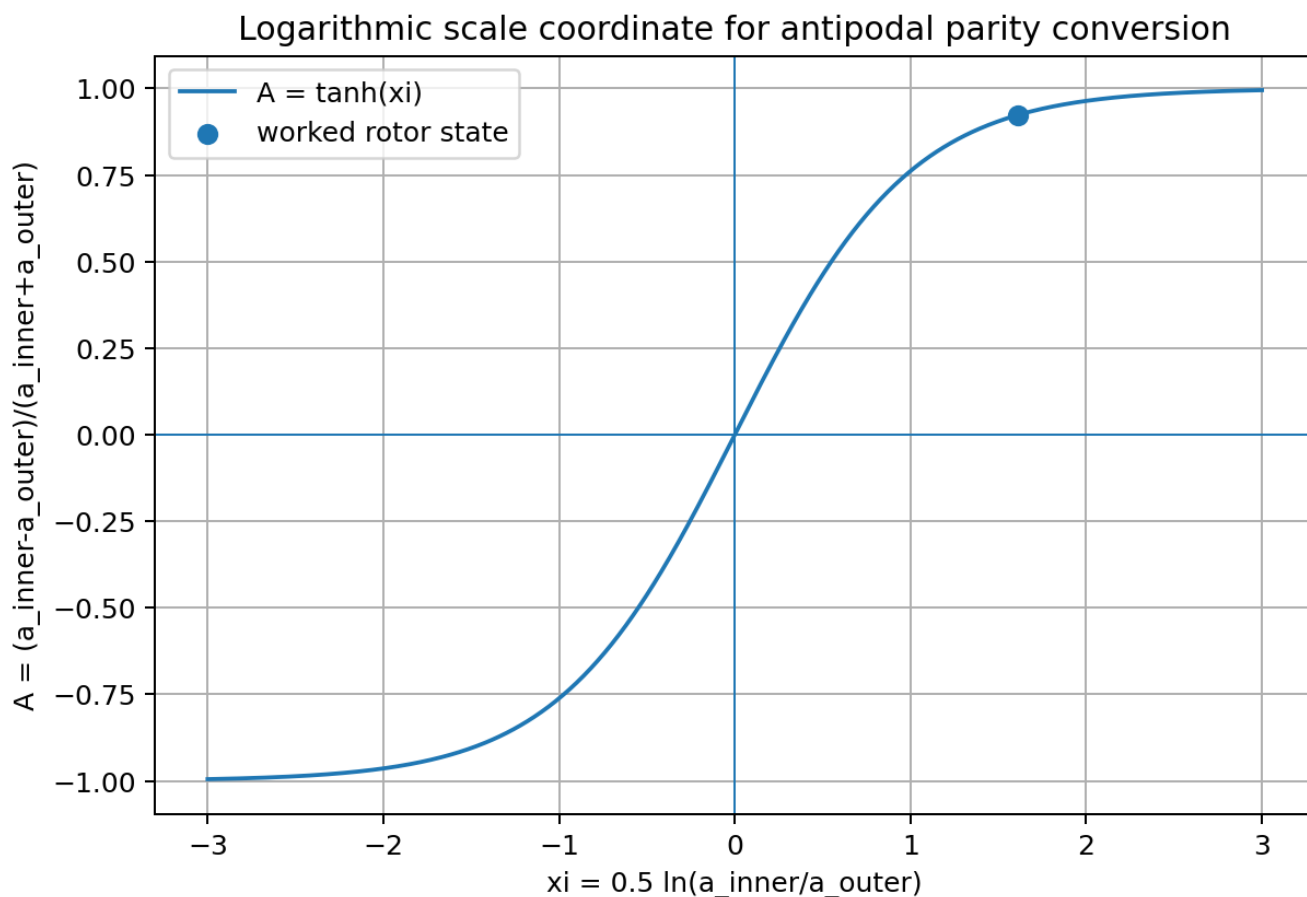


Figure 10. The normalized residual is the hyperbolic tangent of the logarithmic response-weight contrast. The marked point is the worked 5:1 radius-ratio ring pair.

8. Worked concentric-rotor state

Quantity	Inner rotor	Outer rotor
Mass	0.1 kg	1.0 kg
Radius	0.02 m	0.1 m
Shape factor beta	1.0	1.0
Moment of inertia	4.000000000000e-05 kg m ²	0.01 kg m ²
Reaction-only speed	100000.000000 rpm	400.000000 rpm
Signed angular momentum	+0.418879020479 kg m ² /s	-0.418879020479 kg m ² /s
Lateral response	0.007859636501 N	0.000314385460 N opposite
Rotational energy	2193.245422464302 J	8.772981689857 J

The inner response weight is $a_i = 2500 \text{ m}^{-2}$ and the outer response weight is $a_o = 100 \text{ m}^{-2}$. Their ratio is 25. The normalized asymmetry is $24/26 = 12/13$, or 0.9230769231. Thus 92.31 percent of the maximum normalized differential response survives the antipodal cancellation.

Derived quantity	Value
a_i	2500.000000000000 m ⁻²
a_o	100.000000000000 m ⁻²
a_i/a_o	25.000000000000
$\xi = 0.5 \ln(s)$	1.609437912434

Derived quantity	Value
$A=\tanh(\xi)$	$0.923076923077 = 12/13$
Force ratio F_i/F_o	25.000000000000
Net force	0.007545251040721 N
Lateral impulse for source time profile	0.011317876561081 N s

Reaction-constrained antipodal doublet: equal $|L|$, unequal response

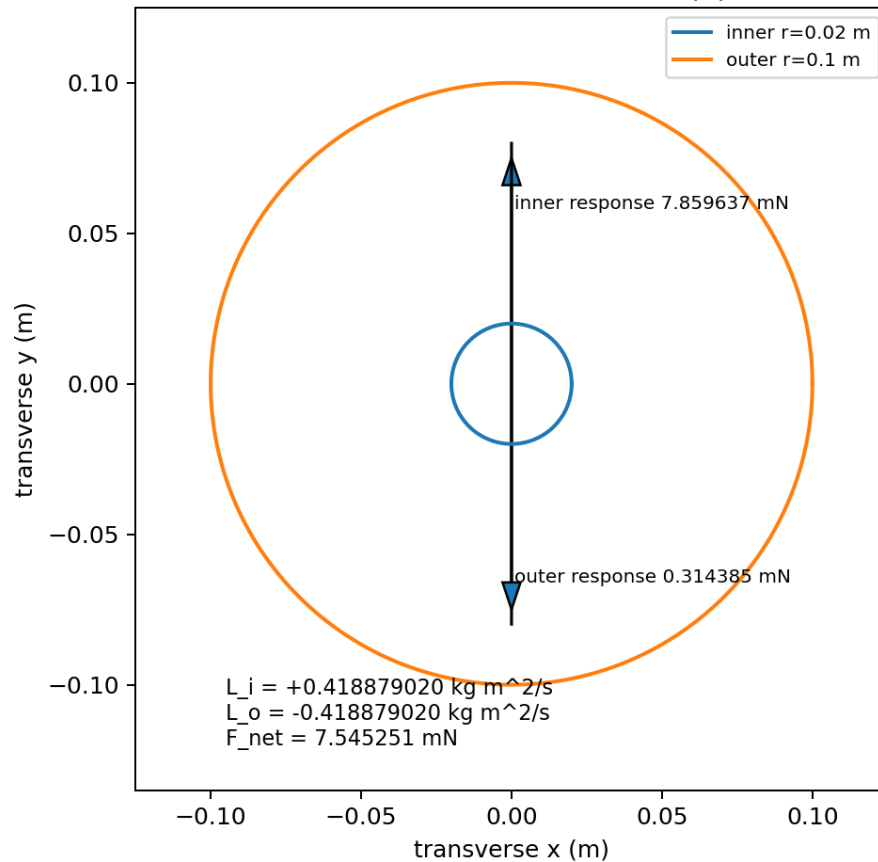


Figure 8. The worked reaction-only state. The two angular momenta cancel exactly, while the force magnitudes differ by the inverse ratio of the specific areal inertias.

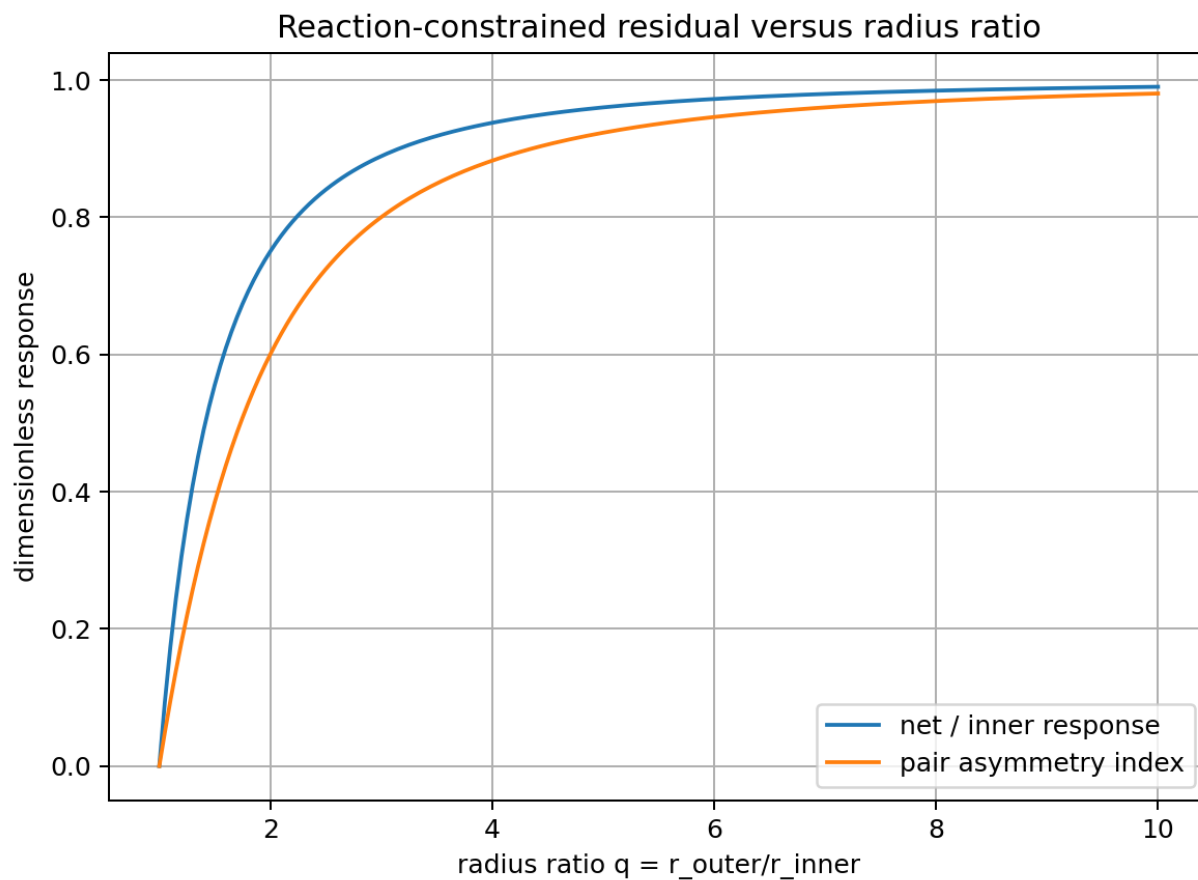


Figure 9. Source-computed radius-ratio sweep. For equal shape factors, the force ratio grows as q^2 , while the normalized residual approaches unity as the radius ratio becomes large.

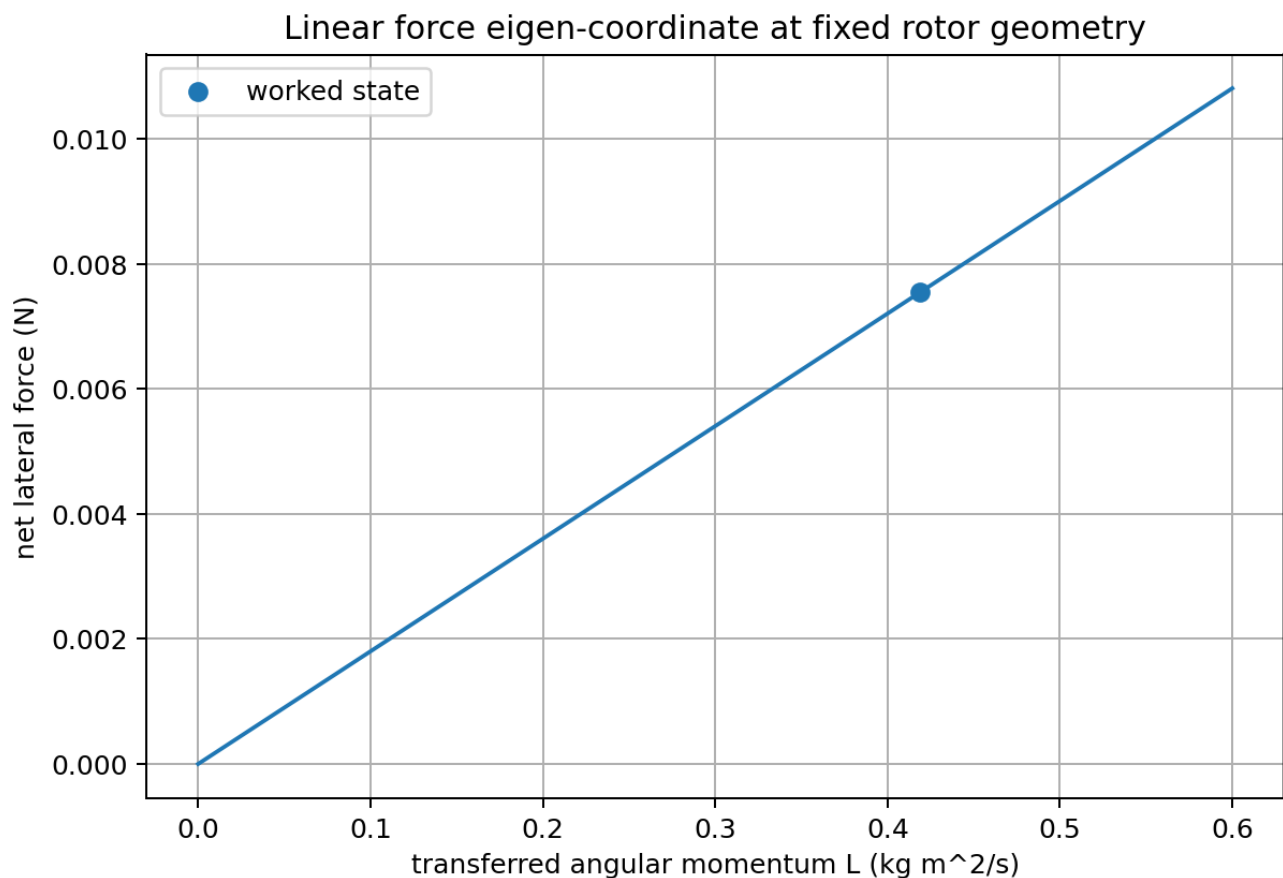


Figure 11. At fixed geometry, the residual force is linear in the transferred angular momentum. The worked state lies on the exact source line.

9. Equal energy, equal angular momentum, and force cancellation are different surfaces

Three balance statements are frequently conflated. Equal rotational energy imposes $I_i \omega_i^2 = I_o \omega_o^2$. Reaction-only angular-momentum balance imposes $I_i \omega_i = I_o |\omega_o|$. Force cancellation imposes $m_i \omega_i = m_o |\omega_o|$. These relations have different powers of radius and mass, so they generally select different operating points.

$$\begin{aligned} \text{equal energy: } \omega_i / |\omega_o| &= (r_o / r_i) \sqrt{\beta_o m_o / (\beta_i m_i)} \\ \text{angular-momentum null: } \omega_i / |\omega_o| &= \beta_o m_o r_o^2 / (\beta_i m_i r_i^2) \\ \text{force null: } \omega_i / |\omega_o| &= m_o / m_i \end{aligned}$$

State	Outer speed	Total angular momentum	Net force
Equal energy	6324.555320337 rpm	-6.204179823385 kg m ² /s	0.002888765916 N
Reaction only	400.000000000 rpm	0.000000000000 kg m ² /s	0.007545251041 N

The equal-energy example requires external angular impulse because its signed total angular momentum is not zero. The reaction-only state starts from zero angular momentum and preserves that constraint, but its energies are strongly unequal. Neither state coincides with the force-null condition for the specified geometry.

10. Mass-Bridge expression of the same separation

The Mass-Bridge archive writes a two-scale counter-rotor model with electron and proton labels. The K-response observable is weighted by $m \omega$, while angular momentum is weighted by $m \rho^2 \omega$. Let $\eta = m_p / m_e$ and $R = \rho_p / \rho_e$. The three balance conditions are:

$$\begin{aligned} \text{K-response null: } \omega_e/|\omega_p| &= \eta \\ \text{equal energy: } \omega_e/|\omega_p| &= \sqrt{\eta} R \\ \text{angular-momentum null: } \omega_e/|\omega_p| &= \eta R^2 \end{aligned}$$

At the reciprocal-Compton placement $R=\eta$, the three frequency ratios are η , $\eta^{3/2}$, and η^3 . Their large separation is not an inconsistency; it records the different powers of radius carried by the three observables.

Condition	At $R=\eta$
K-response null	1836.152673426
Equal energy	78679.7885299238
Angular-momentum null	6.19050912311352e+09

Three distinct balance manifolds across mass and radius scale

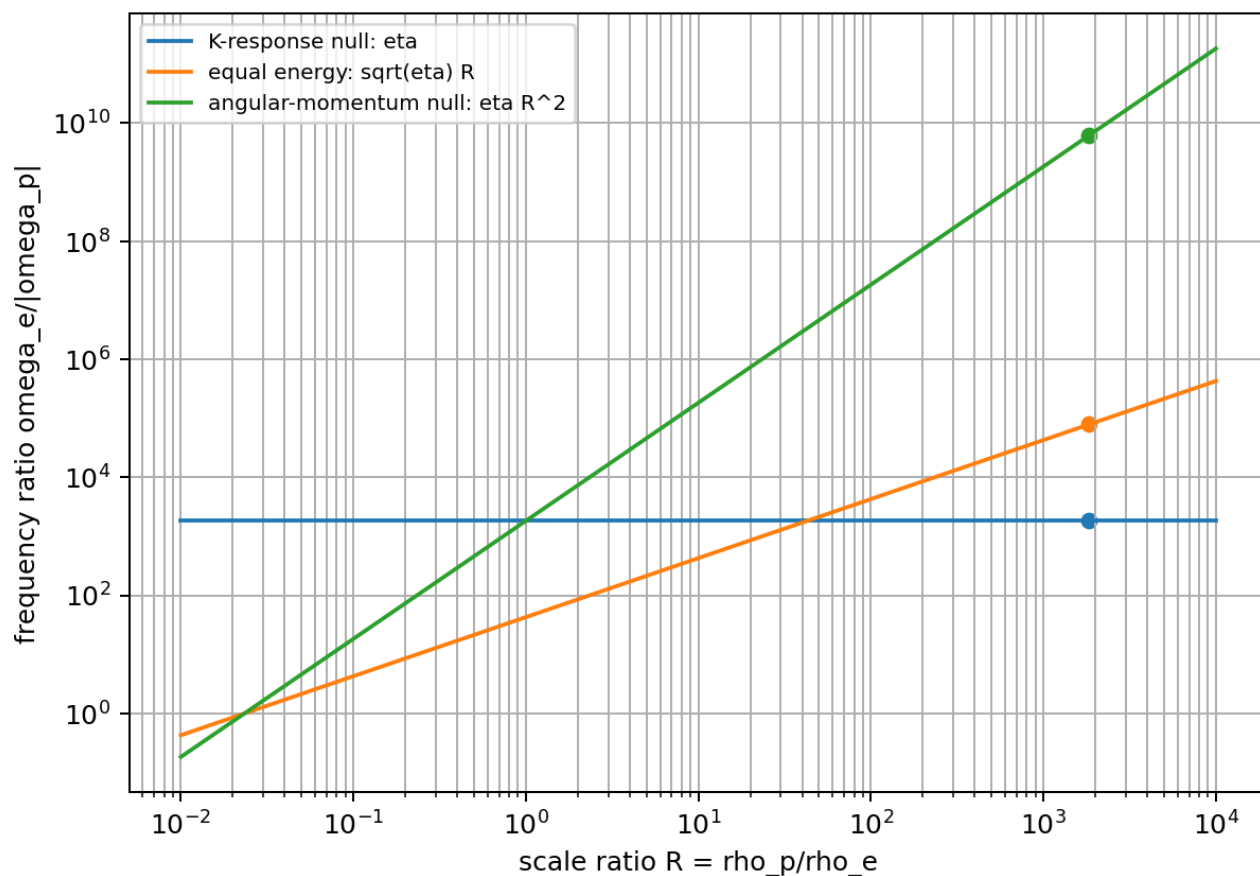


Figure 12. The three Mass-Bridge balance manifolds across six decades of scale ratio. Their differing slopes are the graphical signature of different geometric weights.

10.1 Cell3 face areas as a projective area-ratio system

Bridge quantity	Cell3 value	Known-physics target	Relative residual
1000 A_{bc}/A_{ab}	1836.15268933024	1836.152673426	8.6617200e-09
100 A_{bc}/A_{ac}	137.0360003517588	137.0359991775901	8.5683229e-09
10 A_{ac}/A_{ab}	13.39905342112296	13.39905341987152	9.3397068e-11

The counter-rotor response contrast is also an area-ratio phenomenon because βr^2 is a specific area. The Mass-Bridge uses intrinsic Cell3 face areas; the rotor uses effective inertia areas. The two structures therefore share the same mathematical category: ratios of area-like weights determine which odd states cancel and which project into an even residual.

11. Cell3 metric eigen-dimensions

The symmetric positive-definite Cell3 Gram matrix defines the squared length of every address vector. Diagonalizing it gives three orthogonal principal modes in coefficient space. The eigenvalues have dimensions of native length squared, and their square roots are principal native lengths.

Mode	Eigenvalue native ²	Principal length native	Eigenvector in (a,b,c) coefficient basis
1	0.000417062884363382	0.0204221175289	[0.990721750627, 0.063748853998, -0.120027065482]
2	0.000611511463021504	0.024728757814	[0.117594007608, -0.844848183699, 0.521922595673]
3	0.00316710498583946	0.0562770378204	[-0.068132680917, -0.531194531329, -0.844505954791]

The longest-to-shortest principal-length ratio is 2.755690625. The metric therefore contains a substantial but finite anisotropy. The three metric axes are not the primitive edges; they are eigen-combinations of the primitive directions.

Cell3 metric eigen-dimensions in index space

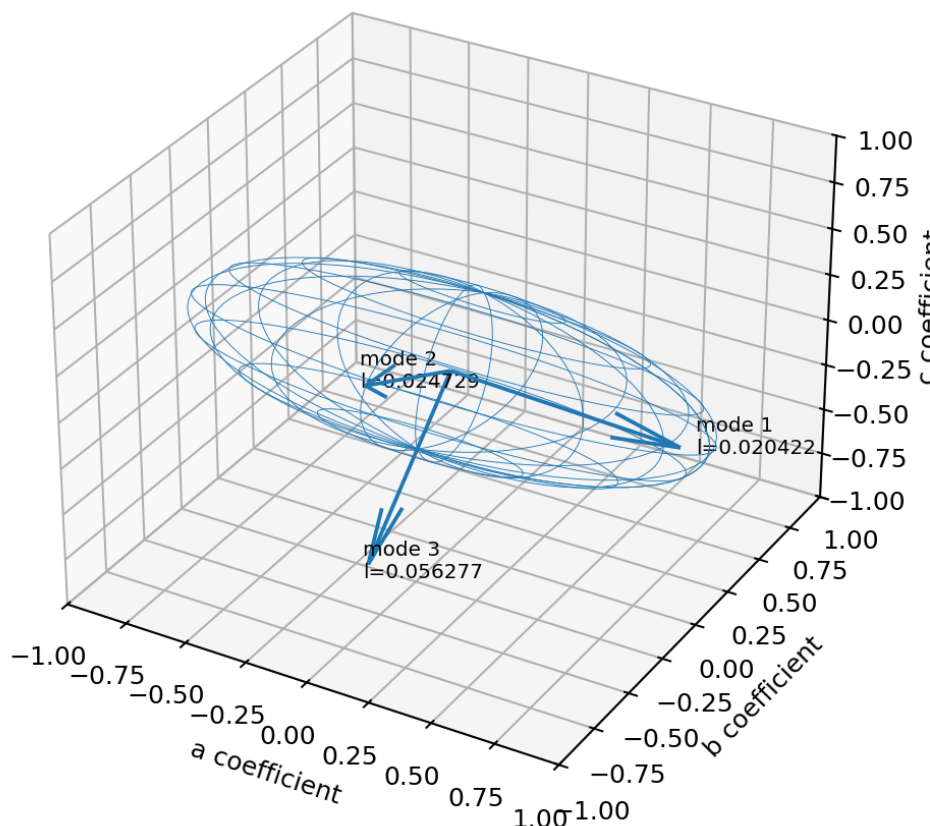


Figure 13. The index-space metric ellipsoid and its three principal eigen-directions. The labels report the square roots of the Gram eigenvalues.

11.1 Three metric axes, two antipodal lifts, and the center

Each principal metric axis has two directed orientations, producing six directed channels. The inversion-fixed center adds one nondirectional channel. This yields a natural 6+1 decomposition. A possible relationship between this finite decomposition and the seven-sheet organization is a structural hypothesis, not a theorem established by the three source archives.

12. Spatial cross-product eigenstructure

The linear map $J_g(\Omega) = \Omega \times g$ is skew-symmetric. Its real nullspace is the one-dimensional span of g . In the transverse plane it acts as a ninety-degree rotation multiplied by $|g|$. Over the complex numbers its eigenvalues are 0 and $\pm i|g|$.

$$\text{eigenvalues}(J_g) = 0, +i \ 930825.217270519235, -i \ 930825.217270519235$$

The physical interpretation is direct. A rotor axis parallel to g produces no lateral response. Any axis in the perpendicular plane produces a force in the same plane, rotated by ninety degrees according to handedness. Reversing the spin changes the sign of Ω and reverses the force.

The complete response architecture can therefore be organized as a product of three independent structures: scale parity determines even or odd exchange character; the Cell3 metric determines preferred internal geometric channels; and transverse helicity determines the spatial direction within the plane perpendicular to g .

$$\text{response space} = \text{scale parity} \times \text{Cell3 metric mode} \times \text{transverse helicity}$$

13. General multirotor covariance theorem

For N rotors, define signed angular momentum $h_j = I_j \omega_j$ and reciprocal specific areal inertia $a_j = 1/(\beta_j r_j^2)$. The total lateral response is the weighted sum of the signed angular momenta.

$$F_{\text{net}}/C_K = \sum_{j=1}^N a_j h_j$$

When total angular momentum is zero, $\sum h_j = 0$. Subtracting any constant mean weight $\bar{a}$ does not change the result.

$$F_{\text{net}}/C_K = \sum (a_j - \bar{a}) h_j$$

Choosing $\bar{a}$ as the arithmetic mean of the a_j gives the covariance form.

Multirotor covariance theorem

For a zero-total-angular-momentum assembly, $F_{\text{net}}/C_K = N \text{Cov}(a, h)$. The residual is the correlation between angular-momentum sign and reciprocal specific areal inertia. Uniform response weight is removed exactly by the angular-momentum constraint.

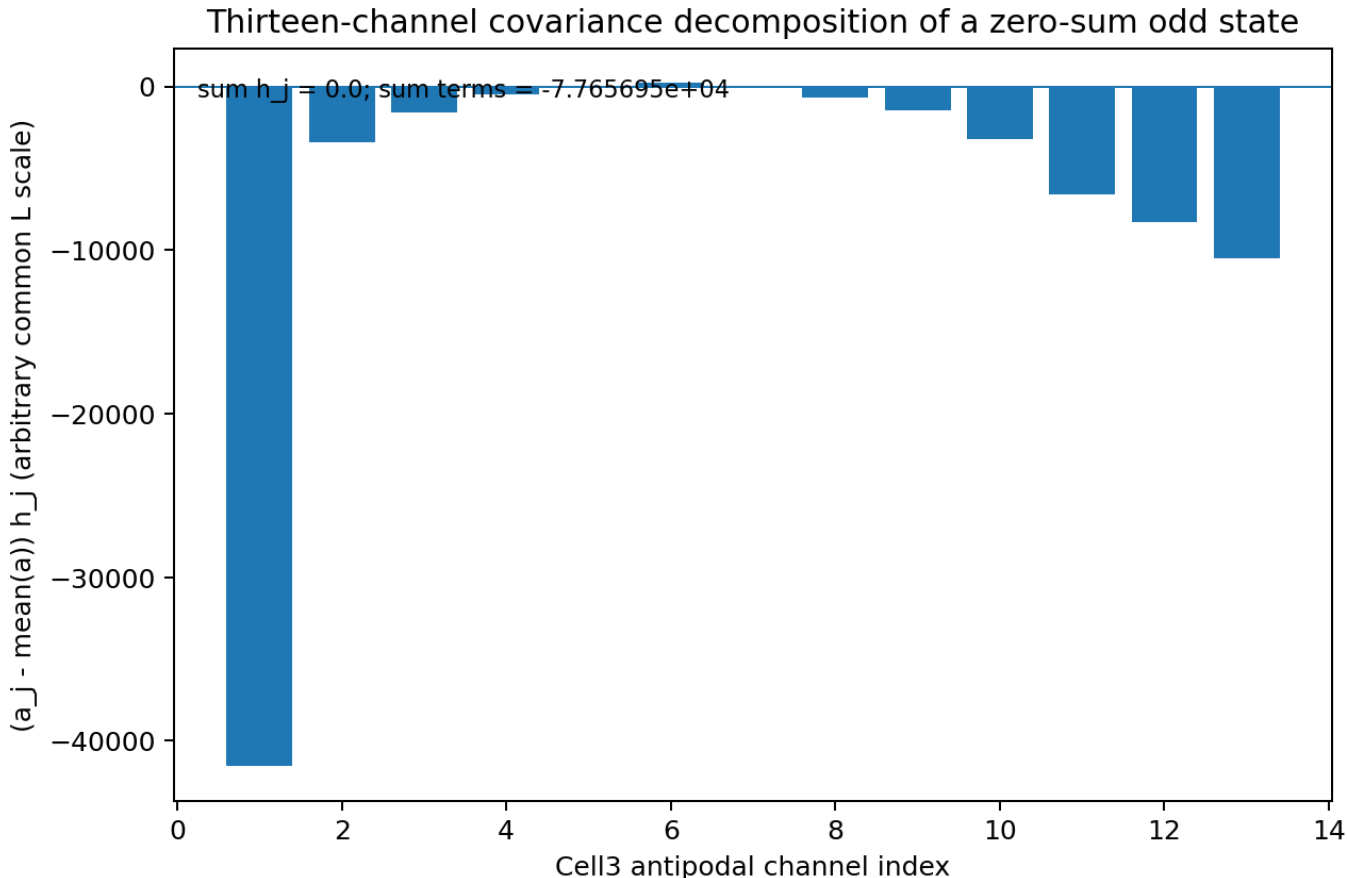


Figure 16. A deterministic 13-channel Cell3 example. The signed h_j values sum to zero, but the channel-by-channel products do not cancel because h_j is correlated with reciprocal pair radius squared.

13.1 Complete 13-dimensional odd Cell3 actuation sector

The 13 antipodal node pairs provide one independent antisymmetric amplitude per pair. Any odd state on the centered 27-node registry can be written as a linear combination of these 13 pair differences. A 13-element coherent architecture can therefore be interpreted as spanning the complete odd inversion eigenspace rather than as an arbitrary element count. This interpretation is a direct mathematical consequence of the finite inversion representation, although its assignment to a physical actuator array is a structural extension beyond the three source files.

14. Design laws and scaling consequences

14.1 Radius contrast at fixed angular momentum

For equal shape factors, $F_i/F_o = (r_o/r_i)^2$. The smaller-radius rotor dominates the response at fixed $|L|$ because it must rotate faster by the same squared radius ratio. The normalized residual approaches one as the radius ratio becomes large.

$$F_i/F_o = \beta_o r_o^2 / (\beta_i r_i^2)$$

$$A = (q^2 - 1) / (q^2 + 1) \text{ for } \beta_i = \beta_o, q = r_o/r_i$$

14.2 Mass changes speed and energy, not fixed-L force

At fixed transferred L and fixed geometry, $\omega = L / (\beta m r^2)$, so increasing mass reduces angular speed. The rotational energy $E = L^2 / (2I)$ also decreases with increasing inertia. The force contribution $C_K L / (\beta r^2)$ remains unchanged. Mass is therefore an engineering variable controlling speed, stress, and stored energy, while the fixed- L force is controlled by geometry.

$$\omega \propto 1/m$$

$$E = L^2 / (2 \beta m r^2)$$

$$F = C_K L / (\beta r^2)$$

14.3 Shape-factor matching

A double-null design does not require equal radii if the shape factors differ. The exact condition is $\beta_i r_i^2 = \beta_o r_o^2$. This permits an experimental null control in which the physical radii remain unequal but the specific areal inertias are matched by changing the mass distribution.

15. Measurement architecture and falsifiable nulls

The mathematical separation suggests a hierarchy of laboratory nulls. Each null changes one weighting relation while preserving the remaining common-mode disturbances. The required observables are rotor speeds, inertias, axis orientation, reaction torque, assembly acceleration or support force, and the field-relative orientation.

Test	Controlled condition	Predicted distinction
Angular-momentum null	$I_i \omega_i + I_o \omega_o = 0$	Residual follows reciprocal specific areal inertia contrast.
Force null	$m_i \omega_i + m_o \omega_o = 0$	Total angular momentum generally remains nonzero.
Double null	$\beta_i r_i^2 = \beta_o r_o^2$ with reaction balance	Both odd angular momentum sum and lateral response cancel.
Spin reversal	$\omega_i, \omega_o \rightarrow -\omega_i, -\omega_o$	Lateral force reverses immediately.
Axis sweep	rotate common spin axis relative to g	Response follows $ \Omega \times g $ and vanishes for parallel alignment.
Radius-ratio sweep	vary r_o/r_i at fixed transferred L	Force ratio follows $\beta_o r_o^2 / (\beta_i r_i^2)$.
Mass substitution	change m at fixed β, r, L	Required speed and energy change; fixed- L force does not.

These tests distinguish geometric parity conversion from ordinary imbalance, bearing drag, motor torque ripple, and support compliance. The strongest internal control is the double-null condition because it preserves counter-rotation while removing the predicted geometric contrast term.

16. Foundational synthesis

The counter-rotating system is an antipodal doublet. Equal and opposite angular momentum places the internal state in an odd exchange eigenmode. If the two channels have the same specific areal inertia, the response operator respects the same exchange symmetry and the lateral response cancels. If their specific areal inertias differ, the response operator contains an odd contrast term that mixes the odd state into an even common-mode force.

antipodal pairing -> odd angular-momentum mode -> scale-dependent parity mixing -> even lateral-force mode

The primary mechanical invariant is not mass, radius, speed, or energy alone. It is the relationship between signed angular momentum and reciprocal specific areal inertia. Mass and velocity determine how an angular-momentum state is realized. Geometry determines how that state is observed by the lateral-response functional.

Cell3 reveals the higher-dimensional organization. Its 27 centered nodes divide into one center and 13 antipodal pairs, producing a 14-dimensional even sector and a 13-dimensional odd sector. Its Gram metric resolves three principal geometric dimensions. Its field response resolves one spatial null axis and two transverse helical modes. Together these structures provide a finite, internally organized address space for counterflow, parity conversion, and directed response.

The multirotor covariance theorem gives the most general compact statement: zero total angular momentum removes only the uniform response mode. It does not remove correlation between angular-momentum sign and geometric scale. That remaining correlation is the residual linear force.

17. New pathways to discovery

17.1 Scale-rapidity composition

Because $\xi = (1/2)\ln(s)$, a hierarchy of nested scale ratios composes by addition. This makes ξ a natural coordinate for cascaded rotor systems, recursive Cell3 constructions, and multiscale phase registries. The corresponding normalized response remains bounded by $|A| < 1$ through the tanh mapping.

17.2 Parity-selective materials and actuators

The odd Cell3 eigenspace suggests designing actuators as antipodal channel pairs rather than as independent elements. Each pair controls one odd coordinate. Geometry or material distribution can then provide the response-weight contrast that maps selected odd coordinates into an even force channel.

17.3 Spectral design of multirotor arrays

For an array, the diagonal response weights and the angular-momentum constraint define a linear algebra problem. Singular or eigenmode analysis can identify the zero-angular-momentum state that maximizes weighted response subject to stress, energy, and speed limits. The two-rotor result is the first nontrivial member of this broader constrained spectral design problem.

17.4 Cell3 metric and parity tensor products

Combining the 13-dimensional odd inversion sector with the three metric eigen-directions and the two transverse helicities produces a structured family of channels. The precise physical selection rules remain a research question, but the mathematical factorization provides a disciplined space in which to formulate and test them.

18. Consolidated conclusions

1. Counter-rotation creates an antipodal odd state but does not by itself guarantee a residual force.
2. At fixed transferred angular momentum, each rotor response is proportional to $1/(\beta r^2)$; mass cancels from the force but remains in speed and energy.
3. Zero total angular momentum and zero lateral response coincide only when the two specific areal inertias βr^2 are equal.

4. In the parity basis, odd angular momentum multiplied by odd geometric contrast produces an even common-direction force.
5. The normalized conversion coefficient is $A=\tanh(x_i)$, where x_i is the additive logarithmic scale coordinate.
6. The centered Cell3 registry has exactly 13 odd inversion dimensions and 14 even dimensions.
7. The Cell3 Gram metric has three principal eigen-dimensions with principal-length anisotropy 2.755690625.
8. The spatial response operator has one null axis parallel to g and two transverse helical modes.
9. For N rotors with zero total angular momentum, the residual is $N C_K \text{Cov}(a,h)$.
10. The central principle is geometric correlation: angular-momentum cancellation removes the uniform mode but leaves scale-correlated odd structure available for conversion into linear response.

Appendix A. Complete numerical source and derived constant table

Name	Definition	Value	Unit
C_K	rotor response coefficient	7.505400000000000e-06	N s/kg
kappa_F	dimensionless coupling	1.61263357733437e-11	1
g	transformed field magnitude	9.30825217270519e+05	m/s
eta	proton/electron mass ratio	1836.152673426	1
alpha^-1	inverse fine-structure constant	137.03599917759	1
a_i	$1/(\beta_i r_i^2)$	2500	m^-2
a_o	$1/(\beta_o r_o^2)$	100	m^-2
s	a_i/a_o	25	1
xi	$0.5 \ln(s)$	1.6094379124341	1
A	$\tanh(\xi)$	0.923076923076923	1
L	reaction-only transferred angular momentum	0.418879020478639	kg m^2/s
F_net	reaction-only net lateral force	0.00754525104072091	N
lambda_1	smallest Gram eigenvalue	0.000417062884363382	native^2
lambda_2	middle Gram eigenvalue	0.000611511463021504	native^2
lambda_3	largest Gram eigenvalue	0.00316710498583946	native^2
ell_3/ell_1	principal-length anisotropy	2.75569062516767	1

Appendix B. Compact derivation ledger

B.1 Reaction residual

$I_j = \beta_j m_j r_j^2$; $h_j = I_j \omega_j$; $F_j = C_K m_j \omega_j = C_K h_j / (\beta_j r_j^2)$. For $h_i = +L$ and $h_o = -L$, $F_{\text{net}} = C_K L[a_i - a_o]$.

B.2 Parity conversion

$h = \sqrt{2} L e_{\text{minus}}$. $D_{\text{pm}} = [[a_{\text{bar}}, \delta_a], [\delta_a, a_{\text{bar}}]]$. Therefore $D_{\text{pm}} h$ has even component $\sqrt{2} \delta_a L$.

B.3 Scale coordinate

$s = a_i/a_o$. $A = (s-1)/(s+1)$. With $s = \exp(2\xi)$, $A = (\exp(2\xi)-1)/(\exp(2\xi)+1) = \tanh(\xi)$.

B.4 Multirotor covariance

If $\sum h_j = 0$, then $\sum a_j h_j = \sum (a_j - a_{\text{bar}}) h_j$. With arithmetic means, $\sum (a_j - a_{\text{bar}}) h_j = N \text{Cov}(a, h)$.

B.5 Inversion dimensions

The 27-node set consists of one fixed center and 13 two-element orbits. Each orbit gives one even and one odd eigenvector. Thus $\dim \text{even} = 14$ and $\dim \text{odd} = 13$.

B.6 Spatial spectrum

The matrix representing $x \rightarrow x \text{ cross } g$ is real skew-symmetric. Its characteristic polynomial is $\lambda(\lambda^2 + |g|^2)$, yielding 0 and $\pm i|g|$.

Appendix C. Reproducibility record

Role	Filename	SHA-256	Bytes
concentric_rotor_source	K-Field_concentric_rotor_lateral_force_analysis_1785171858(1).json	854bf593256b5ee18ca02a65eb1f202a449266b815314b39fbacbf7588e28673	765870
mass_bridge_source	K-Field_Compton_classical_radius_Mass_Bridge_triangle_1785175483(1).json	9301e9108b98c848a14a188b0858a81a7509b53e02cd5de4a255fe8a0a0ece93	42979
antipodal_theorem_source	K-Native_Cell3_antipodal_sphere_theorem1784492216(3).json	2c6331de64d582f58b287f8bda04ca2cf90e70e7eae1c1d463b5f1ec8cfa5005	14664

The machine-readable companion K-Field_counter-rotating_system_is_an_antipodal_doublet_1785179808.json contains the complete 120-entry assimilation anchor list, the 475-node structural coverage registry, all source hashes, the full 27-node coordinate ledger, the source inputs used in the calculations, every derived scalar reported in this paper, the equation ledger, and the figure manifest.

Appendix D. Figure manifest

ID	Filename	Title	Data basis
1	figure_01_base_cell_8_vertices.png	Dimensionally correct Cell3 base cell with all eight vertices.	Mass-Bridge canonical vectors and vertex sums
2	figure_02_all_27_nodes_canonical.png	All 27 centered Cell3 half-step nodes, labeled by address.	Mass-Bridge centered_half_step_nodes
3	figure_03_study_coordinates_planes_nodes.png	All 27 theorem nodes in locked study coordinates with the central plane family.	Native Cell3 theorem lucas_nodes
4	figure_04_face_ab.png	Dimensionally correct ab face plane with all four vertices, edge lengths, included angle, and area.	Canonical vectors a and b
5	figure_05_face_ac.png	Dimensionally correct ac face plane with all four vertices, edge lengths, included angle, and area.	Canonical vectors a and c
6	figure_06_face_bc.png	Dimensionally correct bc face plane with all four vertices, edge lengths, included angle, and area.	Canonical vectors b and c
7	figure_07_13_antipodal_pairs.png	The exact $27 = 1 + 2(13)$ inversion pairing of the centered Cell3 registry.	Mass-Bridge centered nodes paired under $p \rightarrow -p$
8	figure_08_concentric_antipodal_doublet.png	Reaction-only concentric rotor example with equal and opposite angular momentum and unequal lateral response.	Rotor worked_input and reaction_only_state
9	figure_09_radius_ratio_residual.png	Normalized reaction-constrained residual and parity asymmetry across the source radius-ratio sweep.	Rotor reaction_radius_ratio sweep
10	figure_10_tanh_scale_coordinate.png	The exact hyperbolic-tangent relation between logarithmic scale contrast and normalized residual.	Derived from reciprocal specific areal inertia
11	figure_11_force_vs_L.png	Net reaction-constrained force as a linear function of transferred angular momentum.	Rotor reaction_force_vs_L sweep
12	figure_12_mass_bridge_balance_manifests.png	Force-null, equal-energy, and angular-momentum-null manifolds from the Mass-Bridge counter-rotor equations.	Mass-Bridge counter_rotor balance_conditions
13	figure_13_metric_eigen_dimensions.png	Index-space metric ellipsoid and the three Cell3 principal eigen-directions.	Native theorem Gram metric eigendecomposition
14	figure_14_parity_mixing.png	Two-channel parity-basis response operator showing odd-to-even conversion.	Derived two-rotor parity transformation
15	figure_15_spatial_force_eigenstructure.png	Earth-CMB transformed field direction, transverse plane, and a sample $\Omega \times \mathbf{g}$ response.	Rotor coordinate_and_field_inputs
16	figure_16_multirotor_covariance.png	A deterministic 13-channel zero-sum angular-momentum state showing residual covariance with reciprocal radial scale.	Cell3 antipodal pair radii and multirotor covariance theorem

References

1. K-Field Concentric Counter-Rotating Rotor Lateral-Force Analysis, generation epoch 1785171858, machine-readable source ledger.
2. K-Field Compton-Classical-Radius Mass-Bridge Triangle, generation epoch 1785175483, machine-readable source ledger.
3. Native Cell3 Antipodal Sphere Theorem, publication epoch 1784492216, machine-readable theorem and finite-domain certificate.